Hierarchical Self-Learning Knowledge Inference Based on Markov Random Field for Semantic Segmentation of Remote Sensing Images

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***Abstract*— Semantic segmentation is one of the most important tasks in the field of remote sensing. As the spatial resolution increases, the remote sensing images can capture more detailed information and make hierarchical semantic interpretation possible. However, hierarchical semantic segmentation encounters high heterogeneity not only within the intra-layer classes but also among inter-layer classes. It brings challenges to semantic segmentation methods such as the convolutional neural network (CNN). In this article, a hierarchical self-learning knowledge inference model (HSKIM) based on the Markov random field (MRF) model is proposed for hierarchical semantic segmentation of remote sensing images. The HSKIM model introduces a new framework that integrates the advantages of CNN-based data feature learning and MRF-based hierarchical semantic inference. It contains three modules: data learning module (*D*), inference units generation module (*I*), and self-learning knowledge inference module (*S*). The module *D* uses CNN to learn specific data features layer by layer and extract preliminary geographical objects as the initial results. The module *I* refines the geographical objects using a novel boundary-preservation trick to generate more accurate inference units with clear geographical meaning. The module *S* introduces a hierarchical object-based MRF model to implement semantic inference among intra-layer and inter-layer inference units, guided by the spatial interactions and geographical criteria. This module can self-learn and update the relationship between classes iteratively and provide the final result. Experiments on the GID dataset with hierarchical classes, alongside 12 state-of-the-art CNN-based methods, validate the effectiveness and robustness of the proposed HSKIM model. The code of this article is available at https://github.com/iichengzi/HSKIM.**

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***Index Terms*— Convolutional neural network (CNN), hierarchical semantic inference, hierarchical semantic segmentation, Markov random field model (MRF), remote sensing.**

I. Introduction

SEMANTIC segmentation plays an important role in the field of remote sensing image processing. It aims to partition the image into homogeneous regions that can represent different classes of land covers or land uses. In the past decades, many semantic segmentation methods have been proposed for remote sensing images, such as convolutional neural network (CNN) [1], [2], [3], random forest [4], [5], and Markov random field (MRF) model [6], [7], [8]. They have achieved much progress in terms of both accuracy and efficiency. But the semantic segmentation is still far from being solved. One obstacle factor is that the features captured by existing methods cannot effectively distinguish the homogeneity within each land class and the heterogeneity between different land classes. This brings challenges to extracting geographical objects and classifying them into semantic labels. To handle this issue, the geographic object-based image analysis (GEOBIA) has emerged to use superpixels, i.e., oversegmented regions, as the basic units [9], [10], [11], it can capture the regional features to describe geographical objects. Furthermore, to highlight the homogeneity within one land class, feature representation with multiscale is proposed to represent the image from macro to micro perspectives. Many proposed methods have promoted feature representation with multiscale, especially the encoder–decoder structure in CNN. However, these methods usually focus on the representation of features. In fact, with the increase of the spatial resolution, more detailed information becomes discernible, making hierarchical semantic interpretation available [12], [13], [14]. This indicates the inference relationships between semantic labels in multilayers are equally important.

Hierarchical semantic segmentation is an extension of semantic segmentation. Semantic segmentation aims to allocate each pixel to a single land class. However, hierarchical semantic segmentation seeks to assign each pixel to hierarchical land classes representing semantics across multiple layers. Fig. 1 shows an instance of hierarchical semantic segmentation with two different layers, where the high layer is used to describe coarse classes, such as the *built-up*, *water*,

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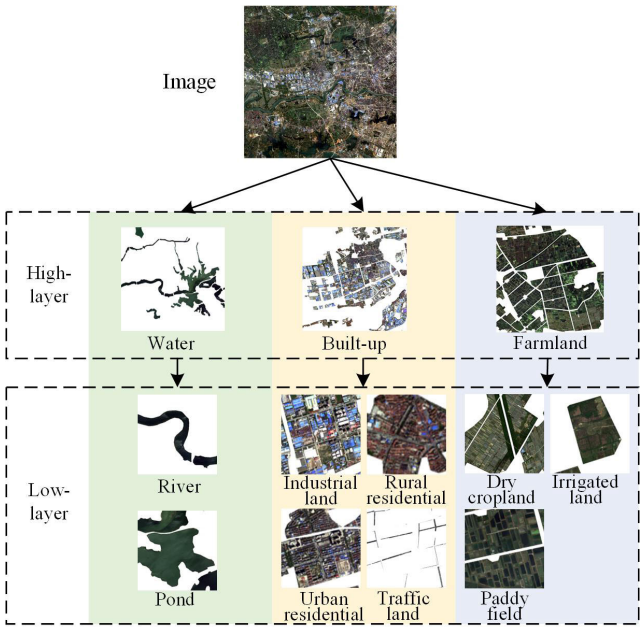


Fig. 1. Geographical objects of hierarchical labels.

and *farmland*. The low layer denotes fine classes, such as the *industrial land*, *rural residential*, *urban residential*, and *traffic land*. Hence, in hierarchical semantic segmentation, beyond feature representation, the inference relationship between different classes of hierarchical layers is also a crucial factor. To model the relationship between hierarchical classes, many methods have been developed. Knowledge graph [15], [16] is one commonly used method to build semantic representation with a hierarchical structure. It represents the information with entities at different levels as nodes and connects these nodes to form a structured knowledge network, which is beneficial for extracting and organizing knowledge from diverse and heterogeneous remote sensing data. However, the inference relationship between these nodes or knowledge still needs an effective and theoretical way to strengthen its interpretability.

MRF, as a probability graph model, has a solid theoretical basis. It can represent and connect geographic knowledge in the probability graph by vertexes and edges, and realizes the feedback iteration between hierarchical geographic knowledge [17]. Specifically, in these MRF-based models for hierarchical semantic segmentation, they usually first extend the unary-layer label field of the MRF model to the multilayer label field for the representation of hierarchical semantic classes, and then implement the semantic inference along the edges between vertexes of the probability graph, i.e., the spatial context, as shown in Fig. 2. However, during the inference process of the MRF model, the basic inference units are usually directly defined as the vertexes in the probability graph, which are the pixels or objects obtained with superpixel methods [18], [19], [20]. As these inference units often lack a clear geographical meaning [21], they will not effectively match the needs of geoscience knowledge inference and limit accuracy. Hence, inference units with more explicit geographical meaning should be generated and incorporated into the MRF-based semantic inference.

Inspired by these efforts, a desirable model for hierarchical semantic segmentation should meet three essential criteria. First, classes of hierarchical labels should be established on the basis of geographical knowledge and geospatial relationships. Second, the model could effectively generate basic units that have a clear geographical meaning and correspond to certain hierarchical labels. Last, it should dynamically integrate geographical knowledge through semantic inference among these hierarchical basic units. Inspired by the above ideas, a hierarchical self-learning knowledge inference model (HSKIM) is proposed in this article. The HSKIM contains three modules, i.e., data learning module (***D***), inference units generation module (***I***), and self-learning update module (***S***), as shown in Fig. 3. Specifically, the module ***D*** initially constructs hierarchical labels following the class guidance of certain geographical criteria. Subsequently, it employs a layer-wise data learning approach with CNN to capture the initial objects corresponding to each semantic label. The module ***I*** introduces a new superpixel-based method to optimize the boundaries of initial objects in the first module. These refined object-based basic units serve as the inference units in this study for further semantic inference. The final module ***S*** proposes a new object-based MRF model to realize semantic inference for integrating hierarchical knowledge. It uses the MRF-based probability graph to capture the spatial interactions between intra-layer and inter-layer inference units and develops a self-learning trick to adaptively learn the relationship among classes of hierarchical layers.

The main contributions of this article lie in the following aspects.

1) This article presents a new hierarchical semantic segmentation framework that can effectively extract geographical objects from the observed image and enable the hierarchical semantic inference among them through the introduction of a novel self-learning technique guided by geographical knowledge.
2) A novel boundary-preservation technique is introduced to generate inference units that correspond to geographical objects. This technique enhances the initial CNN-based results with more accurate boundaries by merging superpixels from the raw image, thereby capturing each geographical object more effectively.
3) A novel MRF-based self-learning technique is proposed to dynamically implement inference among both intra-layer and inter-layer inference units, guided by spatial interactions and geographical relationships. It is designed to merge hierarchical semantic knowledge to produce the final result.

The rest of this article is organized as follows. Section II briefly reviews the related works. Section III introduces the framework of the proposed method and the detailed information of three modules. Section V analyzes the HSKIM model from multiple perspectives and evaluates its effectiveness on GID dataset. Finally, Section V concludes this article.

II. Related Work

Various methods have been studied for semantic segmentation and hierarchical semantic segmentation of high spatial

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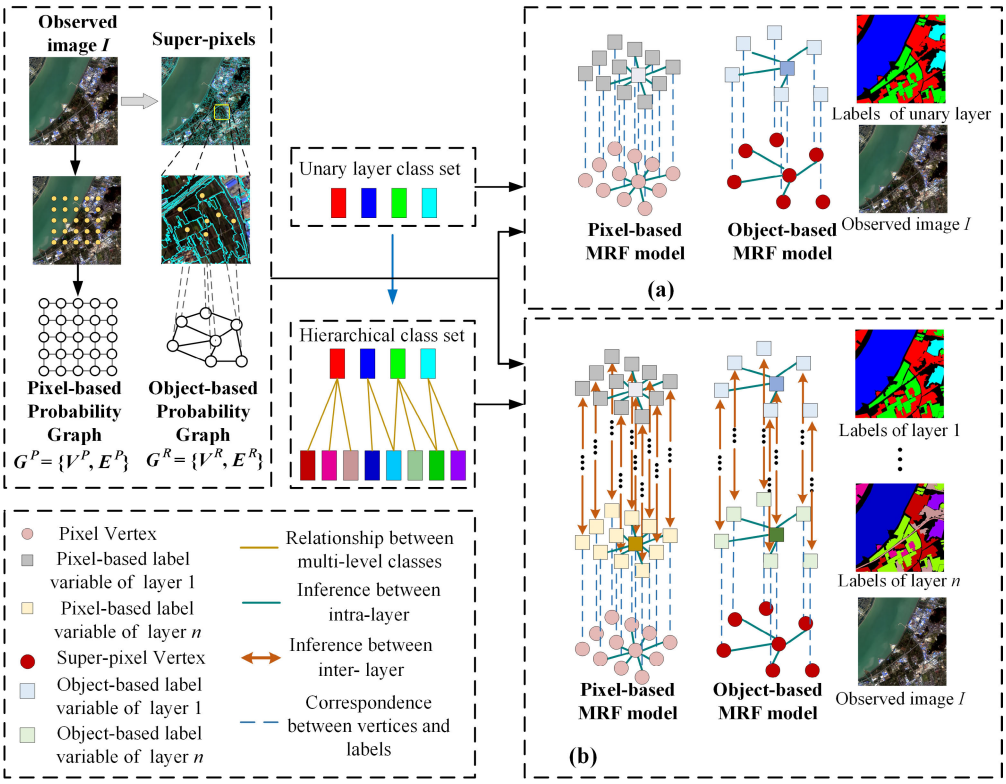


Fig. 2. MRF model with hierarchical semantic layers. (a) Classic MRF model. (b) MRF model with multiclass layers.

resolution remote sensing images. Semantic segmentation usually focuses on feature representation with a single semantic layer, while hierarchical semantic segmentation addresses both the extraction of features and inference with multilayer labels. The related works of these two semantic segmentation tasks are summarized as follows.

A. Semantic Segmentation With Feature Representation

Feature representation is crucial for semantic segmentation, and many methods have been proposed to focus on effective feature representation against the high spatial resolution. Early of these methods are based on pixels, however, as the spatial resolution increases, pixel-based methods struggle to describe the high heterogeneity feature of one land class, and it brings noise results to the semantic segmentation. To overcome this problem, the GEOBIA has been developed to extract regional features of geographical objects. Ming et al. [10] proposed a statistics-based scale parameter selection method for object-based information extraction from high spatial resolution remote sensing images, this is practically helpful and efficient in GEOBIA. Tuia et al. [22] designed a bipartite graph structure to find agreement between probabilistic decisions with multiple spatial supports, which explored the object-based spatial information fused with pixel-based information within an energy minimization framework. Olding et al. [23] proposed an MRF model-based GEOBIA and employed the wavelet transform to get a multiresolution pyramid structure. These methods have enhanced in describing geographical objects with large-scale features. However, GEOBIA may lose some detailed information of geographical objects and result in blurred boundaries. To address this issue, multiscale feature representation has been studied to capture information from macro to detail, such as wavelet transform, quadtree, and CNN. They have shown the effectiveness in multiscale feature representation, especially CNN with its convolution and deconvolution structure [24], [25], [26]. Nie et al. [27] proposed a cross-scale graph interaction network (CGIN) to capture the cross-scale semantic correlation of basic objects and utilized atrous convolutions with different dilation rates to extract multiscale boundary features. Wang et al. [28] designed a boundary-guided fusion module to incorporate the boundary guidance into the multilevel feature fusion operations and combined multilevel features by a geometric-similarity-based boundary loss function. Wang et al. [29] proposed a multiscale enhancement network (MEN), and it used multiscale features of remote sensing images to enhance the network's reconstruction capability. Wu et al. [30] proposed a CNN and multiscale transformer fusion network (CMTFNet) structured with encoder–decoder to extract and fuse local information and multiscale global contextual information of high-resolution remote sensing images.

These methods have promoted the development of semantic segmentation and improved the model's ability to recognize geographical objects. However, they usually focus on the feature representation of geographical objects. As the spatial resolution of remote sensing images increases, more detailed semantic information can be observed, making hierarchical semantic segmentation available. Relationships between hierarchical classes can be involved to assist semantic segmentation of high spatial resolution remote sensing images.

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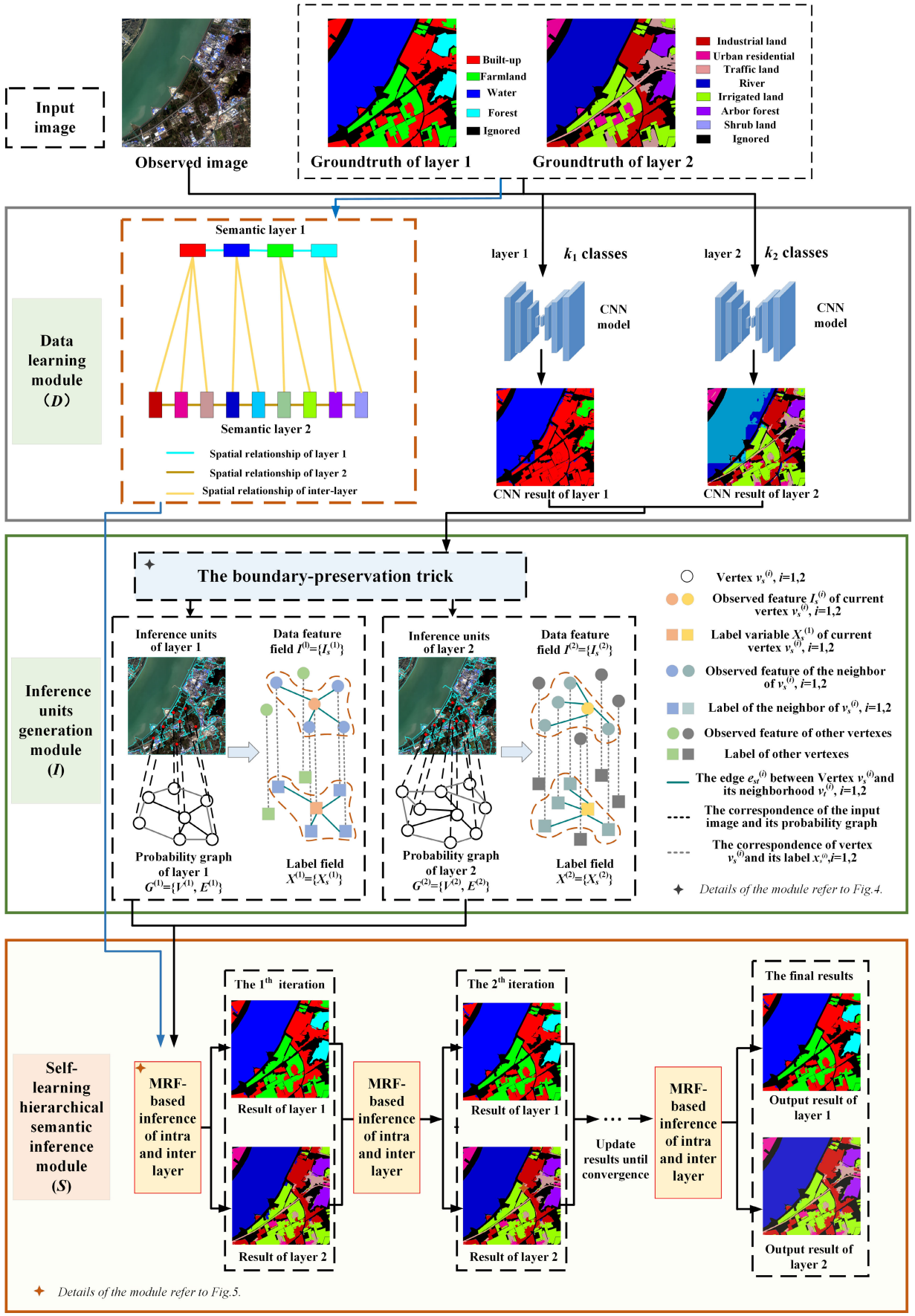


Fig. 3. Flowchart of the proposed HSKIM model, composed by the data learning module (***D***), inference units generation module (***I***), and self-learning hierarchical semantic inference module (***S***). Details of module ***I*** and module ***S*** refer to Figs. 5 and 6.

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B. Hierarchical Semantic Segmentation With Knowledge Inference

Hierarchical semantic segmentation is proposed to consider both the feature representation and hierarchical classes inference. The definition of distinct land classes and the representation of their relationships with hierarchical layers are crucial factors in hierarchical semantic segmentation. Knowledge graph provides a feasible way to construct geographical knowledge and their relationships. Wang et al. [31] proposed a knowledge graph from the geographic perspective, i.e., geographic knowledge graph (GKG), to analyze remote sensing images with hierarchical layers. It expands the concept of the geographic semantic knowledge and classifies semantic into hierarchical layers. Li et al. [32] designed a tree-shaped hierarchical knowledge graph to capture general-to-specific relations between classes for semantic segmentation. Liu et al. [33] constructed a multisource knowledge graph to detect marine oil spills in remote sensing images, combining rule inference with graph neural network technology. Li et al. [34] proposed a model to facilitate knowledge transfer among rich semantic information. It achieves segmentation with hierarchical semantic capability. Tong et al. [35] developed a hybrid classification method by combining patch-wise classification and hierarchical segmentation and created a large-scale land cover dataset for CNN pre-training. All these methods improve the expression of hierarchical semantics for various applications. However, they are usually embedded into networks, causing a lack of sufficient and theoretical interpretability. Thus, it becomes imperative to develop an effective and theoretical inference trick for appropriately integrating the hierarchical semantic information within the intra-layer and inter-layer.

MRF model can not only effectively describe the spatial contextual information but also realize the inference between hierarchical semantic layers with the conditional probability, and it has a solid theoretical basis. Ladicky et al. [36] proposed an object-based associative hierarchical MRF model that used graph-cut based move-making algorithm to achieve the inference between labels and auxiliary labels. Kumar and Hebert [37] built a two-layer hierarchical random field model to exploit the inference between contextual information of different layers, where one auxiliary layer was inferred with a pixel-based short-range context and another label layer was inferred with the object-based long-range context. The authors of this article also proposed object-based MRF models with two-layer and three-layer label fields to realize semantic inference and integrate hierarchical information between the high-layer classes and the low-layer classes [38], [39]. Furthermore, a hybrid MRF model was built on both pixel and object granularities [40]. It proposed a generative alternating granularity inference module within the hybrid probability graph. These MRF-based methods explore an implementable way to capture the inference relationship between hierarchical layers and provide a theoretical basis for the inference between the intra-layer and the inter-layer from the random field perspective. But their inference units usually constructed by superpixel methods, which make the units lack clear geographical meanings.

How to extract hierarchical inference units with more explicit geographical meaning is still a challenging task in hierarchical semantic segmentation for remote sensing images. In the proposed method, the HSKIM model designs a new trick to generate inference units with more clear geographical meanings and complete hierarchical semantic inference within the intra-layer and inter-layer.

III. Proposed Method

The proposed HSKIM model aims to establish hierarchical labels based on geographical knowledge and to implement semantic inference between intra-layer and inter-layer labels. It contains three modules, as illustrated in Fig. 3. In this section, each module of this HSKIM will be discussed in detail.

A. Data Learning Module

This module involves two tasks. One task is to construct classes of hierarchical labels based on specific geographical criteria. In this article, the Gaofen image dataset (GID) [35], [41] is employed as the experimental dataset, aligning with the Chinese Land Use Classification Criteria (GB/T21 010-2017) and containing two-layer hierarchical labels, as shown in Fig. 4(a). Please note that this geographical criteria can be flexibly replaced with other criteria for various applications. The other task is responsible for learning data features of each label and producing an initial segmentation result using the CNN method. Specifically, as land use classes within different layers have different appearances, to learn the specific data features of each layer from the observed image I, the CNN method is used to train the model layer by layer and capture the layer-wise features. These CNN results are also used as the initial results $Y^{(l)}$, as shown in Fig. 4(b). Here, $l \in \{1, 2\}$ denotes the l-layer of hierarchical labels.

B. Inference Unit Generation Module

This module is developed to generate inference units by extracting geographical objects from the initial results $Y^{(l)}$ and construct hierarchical graph models among these inference units based on their spatial interaction. When generating the inference units, a new superpixel method is proposed to catch each geographical object by merging the observed image I and the initial CNN results $Y^{(l)}$. First, it employs a superpixel method $\Gamma(\cdot)$ to, respectively, divide I and $Y^{(l)}$ into oversegmented superpixels, i.e., $\Gamma(I) = \boldsymbol{R}^{\boldsymbol{I}} = \{R^I_{s'} | s' \in \boldsymbol{S}^{\boldsymbol{I}}\}$ and $\Gamma(Y^{(l)}) = \boldsymbol{R}^{\boldsymbol{Y^{(l)}}} = \{R^{Y^{(l)}}_{t'} | t' \in \boldsymbol{S}^{\boldsymbol{Y^{(l)}}}\}$. Here, each $R^I_{s'}$ and $R^{Y^{(l)}}_{t'}$ denote a superpixel obtained from I and $Y^{(l)}$ by $\Gamma(\cdot)$, s' and t' are the sites of corresponding superpixels, respectively. $\boldsymbol{S}^{\boldsymbol{I}}$ and $\boldsymbol{S}^{\boldsymbol{Y^{(l)}}}$ are the sets of sites s' and t', $\boldsymbol{R}^{\boldsymbol{I}}$ and $\boldsymbol{R}^{\boldsymbol{Y^{(l)}}}$ are the sets of $R^I_{s'}$ and $R^{Y^{(l)}}_{t'}$. Second, as shown in Fig. 5(a), each superpixel $R^I_{s'}$ within the set $\boldsymbol{R}^{\boldsymbol{I}}$ can effectively identify a portion of the object's boundary, but it typically represents only a small fragment of one geographical object. Conversely, each $R^{Y^{(l)}}_{t'}$ can roughly represent the entire geographical object but may suffer potential inaccuracies in identifying the object's boundary,

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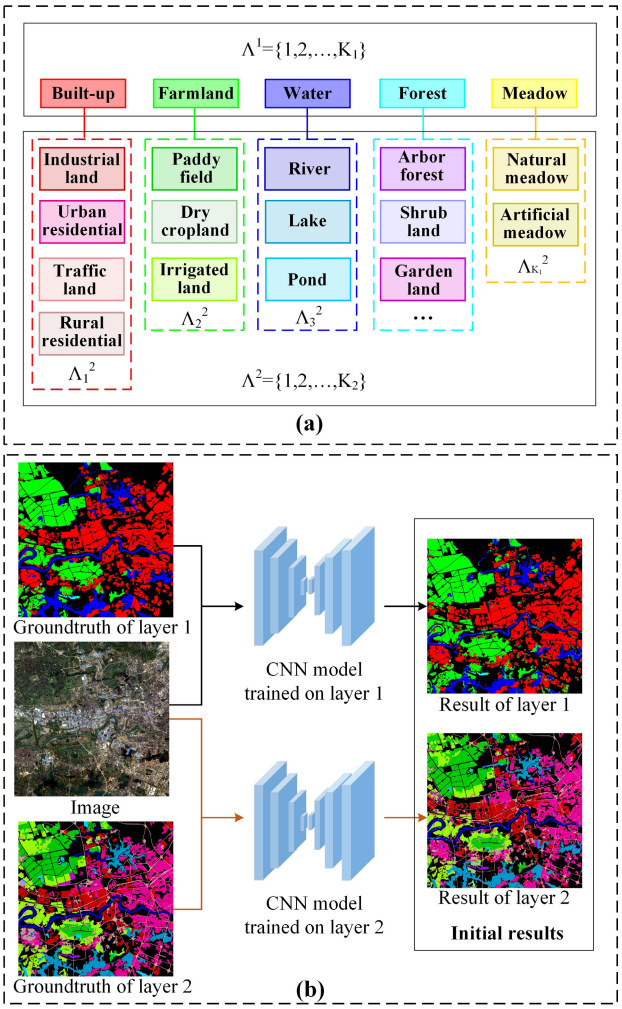


Fig. 4. Land use classification criteria employed in this article and examples of the data learning module. (a) Land use classification criteria employed in this article. (b) Examples of the data learning module.

as shown in Fig. 5(b). To combine the advantages of both $\boldsymbol{R}^{\boldsymbol{Y}^{(l)}}$ and $\boldsymbol{R}^{\boldsymbol{I}}$, the superpixel method in this section develops a new trick comprising three steps: boundary searching, boundary determining, and boundary refining, as shown in Fig. 5. During the boundary searching step, for each superpixel $R_{t'}^{Y^{(l)}}$, if $\exists s' \in S^I$ and $R_{s'}^I \cap R_{t'}^{Y^{(l)}} \neq \emptyset$, then these $R_{s'}^I$ are identified as boundary candidates of $R_{t'}^{Y^{(l)}}$. In the boundary determining step, if boundary candidates $R_{s'}^I$ satisfy condition (1)

$$\left|R_{s'}^I \cap R_{t'}^{Y^{(l)}}\right| > \left|R_{s'}^I \cap R_{w'}^{Y^{(l)}}\right| \quad \forall w' \in \boldsymbol{S}^{\boldsymbol{Y}^{(l)}}/t' \tag{1}$$

then these $R_{s'}^I$ are gathered as the boundaries of $R_{t'}^{Y^{(l)}}$. In the last step, boundary refining is implemented according to the following criterion:

$$R_s^{(l)} = \bigcup_{s'} R_{s'}^I. \tag{2}$$

This generated superpixel $R_s^{(l)}$ could effectively extract geographical objects with the boundary preservation. It is referred to as the inference unit in this article.

Let each inference unit $R_s^{(l)}$ denote a vertex $v_s^{(l)}$ in a graph, the graph model of layer l can be constructed as $G^{(l)} = \{V^{(l)}, E^{(l)}\}$, $l \in \{1, 2\}$. Here, $V^{(l)} = \{v_s^{(l)} | s \in \boldsymbol{S}^{(l)}\}$ is the vertex set and $\boldsymbol{S}^{(l)}$ is the site set. $E^{(l)} = \{e_{s,t}^{(l)} | s, t \in \boldsymbol{S}^{(l)}\}$ is the edge set. Each edge $e_{s,t}^{(l)}$ represents the spatial interaction between vertex $v_s^{(l)}$ and vertex $v_t^{(l)}$, namely $e_{s,t}^{(l)} = 1$ if vertex $v_s^{(l)}$ and vertex $v_t^{(l)}$ are adjacent, $e_{s,t}^{(l)} = 0$ if they are not.

C. Self-Learning Hierarchical Semantic Inference Module

This module is proposed to achieve semantic inference among intra-layer and inter-layer inference units and provide the final result by integrating their hierarchical information. Because each inference unit generated in the preceding module has a relatively clear geographical interpretation, the main challenge of this module lies in incorporating geographical relationships to guide semantic inference. To address this issue, a self-learning object-based MRF model is introduced for hierarchical semantic inference in this section. The proposed HSKIM model first defines the feature fields $I^{(l)} = \{I_s | s \in S^{(l)}\}$ and the label fields $X^{(l)} = \{X_s^{(l)} | s \in S^{(l)}\}$ on graph models $G^{(l)}$. Each $I_s^{(l)}$ represents the feature vector of vertex $v_s^{(l)}$, and each random variable $X_s^{(l)}$ denotes the class of vertex $v_s^{(l)}$, with $X_s^{(l)}$ taking values from Λ^l, $l \in \{1, 2\}$. Here, $\Lambda^1 = \{1, 2, \ldots, K_1\}$ and $\Lambda^2 = \{1, 2, \ldots, K_2\}$ are the class sets of high and low semantic layers, $K_1 < K_2$. To describe the affiliation relationship between classes of high and low layers, Λ^2 is further divided into K_1 subsets Λ_k^2, i.e., $\Lambda^2 = \bigcup_{k=1}^{K_1} \Lambda_k^2$, where if a low-layer class belongs to Λ_k^2, then the high-layer class it is affiliated with is k, $k \in \Lambda^1$, as shown in Fig. 3(a). Let $x^{(l)} = \{x_s^{(l)} | s \in \boldsymbol{S}^{(l)}\}$ denote the realizations of $X^{(l)} = \{X_s^{(l)} | s \in \boldsymbol{S}^{(l)}\}$. In the HSKIM model, the semantic segmentation problem of each layer l can be converted to find the optimal realization $\tilde{x}^{(l)}, l = 1, 2$, that is,

$$\begin{aligned}
\tilde{x}^{(l)} &= \arg\max_{x^{(l)}} P\left(X^{(l)} | I\right) P\left(X^{(l)} | X^{(3-l)}\right) \\
&= \arg\max_{x^{(l)}} P\left(I | X^{(l)}\right) P\left(X^{(l)}\right) P\left(X^{(l)} | X^{(3-l)}\right) \\
&= \arg\min_{x^{(l)}} -\ln\left\{P\left(I | X^{(l)}\right) P\left(X^{(l)}\right) P\left(X^{(l)} | X^{(3-l)}\right)\right\} \\
&= \arg\min_{x^{(l)}} \varphi\left(X^{(l)}, I\right) + \psi\left(X^{(l)}\right) + \xi\left(X^{(l)} | X^{(3-l)}\right).
\end{aligned} \tag{3}$$

In the above function, $\varphi(X^{(l)}, I)$ describes the feature information of the observed image, $\psi(x^{(l)})$ and $\xi(x^{(l)}, x^{(3-l)})$ represent the intra-layer and inter-layer class relationships. The proposed MRF model designs a probability way to integrate data features and hierarchical classes when finding the optimal realization. The detailed definitions of $\varphi(X^{(l)}, I)$, $\psi(x^{(l)})$ and $\xi(x^{(l)}, x^{(3-l)})$ are given as follows.

$\boldsymbol{\varphi(X^{(l)}, I)}$: It is the likelihood function to describe the feature information of I. For each site $s \in S^{(l)}$, this term is defined as the Gaussian distribution conditioned on $X_s^{(l)} = h^{(l)}, h^{(l)} \in \Lambda^l$. That is,

$$\begin{aligned}
\varphi(X_s^{(l)} = h^{(l)}, I) = \frac{1}{2}\Big[&p \cdot \ln(2\pi) + \ln(\Sigma_{h^{(l)}}) \\
&+ \left(I_s^{(l)} - \mu_{h^{(l)}}\right)^T \Sigma_{h^{(l)}}{}^{-1} \left(I_s^{(l)} - \mu_{h^{(l)}}\right)\Big]
\end{aligned} \tag{4}$$

where p is the feature dimension of image I, $\mu_{h^{(l)}}$ and $\Sigma_{h^{(l)}}$ represent the mean and variance of class $h^{(l)}$.

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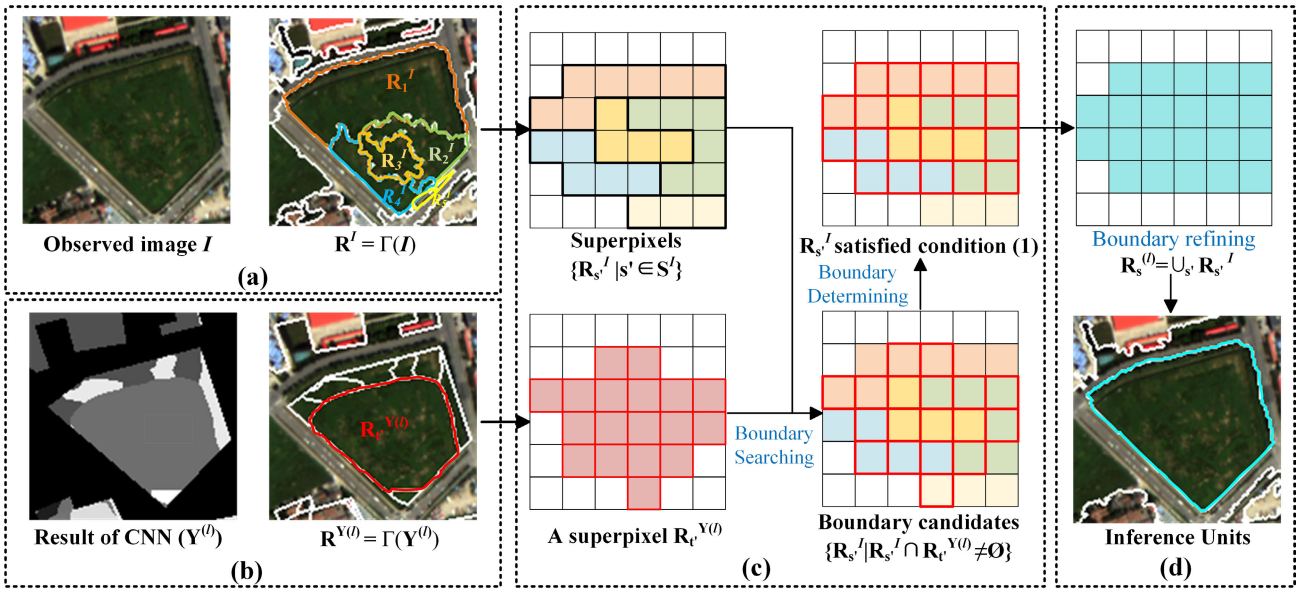


Fig. 5. Flowchart of the boundary-preservation trick. (a) Superpixels generated by observed image. (b) Superpixels generated by CNN-based results. (c) Process of the boundary-preservation trick. (d) Superpixels generated by the proposed trick.

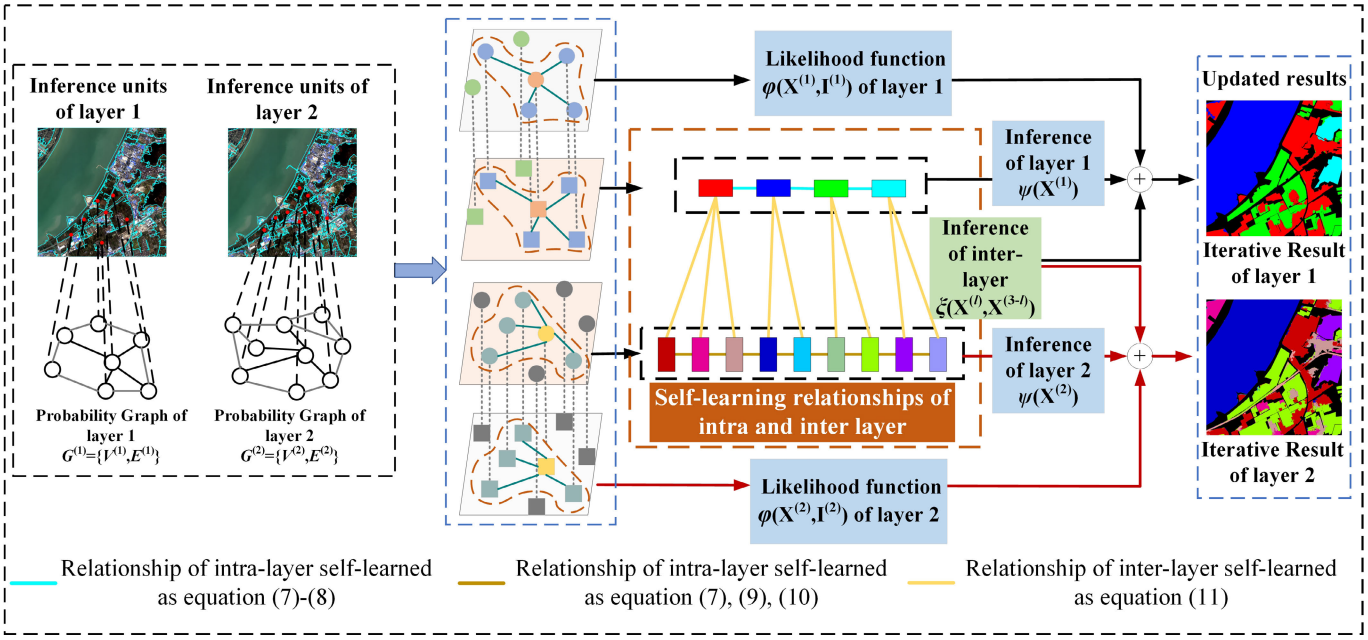


Fig. 6. Flowchart of MRF-based hierarchical semantic inference with self-learning.

$\boldsymbol{\psi}(\boldsymbol{X}^{(l)})$: It is used to model the interaction of inference units within the same layer, which is,

$$\psi(x_s^{(l)}) = \sum_{t \in N_s^{(l)}} V\left(x_s^{(l)} = i, x_t^{(l)} = j\right) \tag{5}$$

where

$$V\left(x_s^{(l)} = i, x_t^{(l)} = j\right) = b_{\mathrm{st}} \cdot \beta_{i,j}^{(l)}, \quad l = 1, 2. \tag{6}$$

Here, $N_s^{(l)} = \{x_t^{(l)} | e_{\mathrm{st}}^{(l)} = 1, t \in S^{(l)}\}$ is the neighborhood set of $x_s^{(l)}$, and b_{st} is the length of the edge shared by region $R_s^{(l)}$ and region $R_t^{(l)}$, describing the strength of the interaction between $R_s^{(l)}$ and $R_t^{(l)}$. And $\beta_{i,j}^{(l)} = c_{i,j}^{(l)} \cdot d_{i,j}^{(l)}$ is the knowledge inference term. It has two parts: $c_{i,j}^{(l)}$ and $d_{i,j}^{(l)}$. The first part $c_{i,j}^{(l)}$ is used to capture the relationship between class i and class j, $i, j \in \Lambda^l$. It can be self-learned as follows, i.e.,

$$c_{i,j}^{(l)} = \begin{cases} \ln \sum\limits_{x_s^{(l)}=i} \sum\limits_{x_t^{(l)}=j, e_{\mathrm{st}}^{(l)}=1} b_{\mathrm{st}}, & l = 1 \\ \sum\limits_{x_s^{(l)}=i} \sum\limits_{x_t^{(l)}=j, e_{\mathrm{st}}^{(l)}=1} b_{\mathrm{st}}, & l = 2. \end{cases} \tag{7}$$

The second part, $d_{i,j}^{(l)}$, is the weight of $c_{i,j}^{(l)}$. When $l = 1$, it is

$$d_{i,j}^{(1)} = \begin{cases} 1, & i \neq j \\ -1, & i = j \end{cases} \tag{8}$$

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when $l = 2$, it is

$$d_{i,j}^{(2)} = \begin{cases} 1, & i \neq j, \text{and } i \in \Lambda_k^2, j \in \Lambda_{k'}^2, k \neq k', k, k' \in \Lambda^1 \\ q, & i \neq j, \text{and } i, j \in \Lambda_k^2, k \in \Lambda^1 \\ -1, & i = j \end{cases} \tag{9}$$

where

$$\begin{aligned} q &= \exp\left\{ P\left(x_m^{(2)} = i | x_n^{(2)} = j, x_m^{(1)} = x_n^{(1)} = k, e_{mn} = 1\right) \right\} \\ &= \exp\left\{ \frac{\left|\{m | x_m^{(2)} = i, x_n^{(2)} = j, x_m^{(1)} = x_n^{(1)} = k, e_{mn} = 1\}\right|}{\left|\{n | x_n^{(2)} = j, x_n^{(1)} = k\}\right|} \right\} \\ & 1 < m, n < M \times N, \quad e_{mn} = 1, i, j \in \Lambda_k^2. \end{aligned} \tag{10}$$

Here, m and n are the site of pixels in observed image I, sized $M \times N$, $x_m^{(l)}$ is the label of pixel m at layer l, and $e_{mn} = 1$ means the region including pixel m is adjacent with the region including pixel n. Please note that, when $i \neq j$, $d_{i,j}^{(l)}$ assigns two different weights for the low semantic layer in (9). This distinction arises from the observation that different low-layer classes belonging to the same high-layer class are more prone to misclassification with each other than those belonging to different high-layer classes. Consequently, the penalized parameter with a high weight, $q > 1$, is introduced to deal with such instances.

$\boldsymbol{\xi(X^{(l)}|X^{(3-l)})}$: It is used to represent the relationship between inter-layer classes. For each site $s \in R^l$, it is defined as

$$\begin{aligned} &\xi\left(x_s^{(l)} = i | j \in \Lambda^{(3-l)}, x^{(3-l)}\right) \\ &\quad = -\alpha^{(l)} \left|\left\{m | x_m^{(3-l)} = j, m \in R_s^{(l)}\right\}\right| \\ &\qquad \cdot \ln P'\left(x_s^{(l)} = i | x_m^{(3-l)} = j\right), \quad j \in \Lambda^{(3-l)} \end{aligned} \tag{11}$$

where

$$\alpha^{(l)} = \ln \frac{\sum_s \left|R_s^{(3-l)}\right|}{\left|\{s | s \in S^{(3-l)}\}\right|} \tag{12}$$

and

$$\begin{aligned} &P'\left(x_s^{(l)} = i | x_m^{(3-l)} = j\right) \\ &\quad = \begin{cases} P\left(x_s^{(l)} = i | x_m^{(3-l)} = j\right), & i \in \Lambda_j^l \\ \left[P\left(x_s^{(l)} = i | x_m^{(3-l)} = j\right)\right]^{\text{iter}}, & \text{other} \end{cases} \end{aligned} \tag{13}$$

with

$$P\left(x_s^{(l)} = i | x_m^{(3-l)} = j\right) = \frac{\left|\left\{m | x_m^{(l)} = i, x_m^{(3-l)} = j\right\}\right|}{\left|\left\{m | x_m^{(3-l)} = j\right\}\right|}.$$

Here, $s \in S^{(l)}$, m is the site of pixels, $x^{(3-l)}$ is the realization of labels at layer $3 - l$, and iter is the number of current iterations. In the above function (11), the global term $P'(x_s^{(l)} = i | x_m^{(3-l)} = j)$ is used to self-learn the relationship between inter-layer classes i and j, it is composed of two forms to conform to the geographical criteria. Please note that, according to the Chinese Land Use Classification Criteria (GB/T21 010-2017) for the GID dataset, the relationship between hierarchical label classes should be a one-to-one correspondence. However, the initial CNN-based results may have misclassifications that do not adhere to this criterion. To deal with these misclassifications, (13) introduces the term $[P(x_s^{(l)} = i | x_m^{(3-l)} = j)]^{\text{iter}}$, which converges to 0 with iteration and aligns with the geographical criteria GB/T21 010-2017. Besides, $\alpha^{(l)}$ is another global term to describe the average size of regions $R_s^{(3-l)}$, it is used to overcome the imbalance of inter-layer relationship caused by the different areas of hierarchical inference units. And the local term $|\{m | x_m^{(3-l)} = j, m \in R_s^{(l)}\}|$ measures the corresponding weight in current region $s \in R^l$.

The flowchart of the proposed MRF-based model is shown in Fig. 6, illustrating the integration of the data feature term $\varphi(X^{(l)}, I)$ and the intra-layer and inter-layer class-relationship terms $\psi(x^{(l)})$, and $\xi(x^{(l)}, x^{(3-l)})$. To self-learn these terms for fitting the specific image, a generative probability inference is proposed to iteratively update and integrate hierarchical data and class information, as shown in the (c) part of Fig. 3.

D. Algorithm of the Proposed HSKIM Model

The algorithm of the HSKIM method is given as follows.

Algorithm 1 Algorithm of the HSKIM Model

Input: The observed image I, the numbers of hierarchical classes K_1 and K_2.

Output: Hierarchical segmentation results based on the HSKIM model.

1: **Data learning module**
Use a CNN model to get the initial results $Y^{(l)}$ layer by layer, $l = 1, 2$.

2: **Inference uints generation module**
Use mean shift provided by EDISON (http://www.wisdom.weizmann.ac.il/bagon/MATLAB.html) and the technique proposed in this article based on equation (1)-(2) to get regional inference units $R^{(l)} = \{R_s^{(l)} | s \in S^{(l)}\}$.

3: **Self-learning hierarchical semantic inference module**
3.0 Initialization Calculate the knowledge relationship $c_{w,t}^{(l)}$ and $d_{w,t}^{(l)}$ based on equation (7)-(9) and the inter-layer balancing parameter $\alpha^{(l)}$. Set $t = 1$, and the realization $x_0^{(l)} = \{x_s^{(l)} | s \in S^{(l)}\}$ is initialized based on $Y^{(l)}$ and $R^{(l)}$.
Solving model with iteration
3.1 Calculate the mean $\mu_{h^{(l)}}$ and variance $\Sigma_{h^{(l)}}$;
3.2 Calculate the relationship of intra-layer by self-learning with equations (7)-(10);
3.3 Calculate the relationship of inter-layer by self-learning with equations (12)-(13);
3.4 Calculate the energy function $\varphi(X^{(l)}, I)$, $\psi(x^{(l)})$, and $\xi(x^{(l)}|x^{(3-l)})$ based on equations (4), (5), (11) and $x_{t-1}^{(l)}$;
3.5 Find the optimal realization $\tilde{x}^{(l)}$ based on equation (3).
3.6 If $\tilde{x}^{(l)} \neq x_{t-1}^{(l)}$, set t=t+1 and go to step 3.1; else, output $\tilde{x}^{(l)}$ as the result.

IV. Experimental Results

The proposed HSKIM model presents a new framework for hierarchical semantic segmentation. To evaluate the effectiveness of the HSKIM model, various types of experiments are

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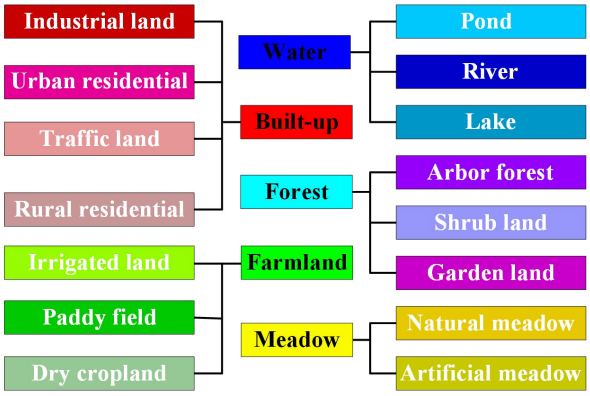


Fig. 7. Land use classification criteria of the GID dataset. Classes of GID-5 are listed on the middle, their subclasses are listed on the sides, with black lines connecting them.

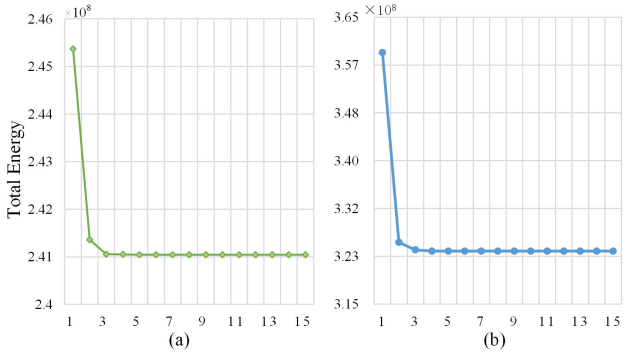


Fig. 8. Energy for each layer. (a) Energy of the high layer. (b) Energy of the low layer.

implemented in this section to test its performance. It involves four sections. First, the test dataset and implementation details are introduced. Second, the HSKIM model is tested from multiple perspectives. It includes the evaluation of hierarchical semantic inference process, the test of the penalized parameter, and the analysis of the ablation study. Third, to test the robustness of the proposed model against different CNN-based methods, the HSKIM model employs 12 CNN-based methods as the baseline for its data learning module. Finally, the performance of the HSKIM model is illustrated and compared with four other MRF-based methods.

A. Dataset and Implementation Details

1) Dataset: To implement hierarchical semantic inference, the GID [35], [41] with two semantic labels is employed. The hierarchical labels follow the Chinese Land Use Classification Criteria (GB/T21 010-2017). The high-layer labels in the GID are categorized into five classes (GID-5) to denote the macroscopic information of geographical objects. GID-5 is further divided into 15 subclasses (GID-15) within five subsets. These low-layer labels are used to show detailed information about geographical objects, as illustrated in Fig. 7. The dataset covers more than 60 different cities, which has a 4-m spatial resolution and sizes of 7200 $\times$ 6800 for each image. There are 130 images in the GID dataset with hierarchical labels, where 120 are used as the training data and the rest ten images are used as the test data.

2) Implementation Details: For all the experiments in both GID-5 and GID-15, we crop the images and corresponding annotation masks into patches of 512 $\times$ 512 pixels, splitting into two subsets, i.e., a total of 25 200 data patches for training and 2100 data patches for testing. The optimizer is set to Adam [42] and the learning rate of CNN-based models is 0.01. The weight decay value is 5e-4. The training epoch is 20 for the dataset, and the batch size is 4. The loss function is set to the cross-entropy [43].

To quantitatively evaluate the performance of the proposed method, three metrics are used to measure experimental results, i.e., Kappa coefficient [44], overall accuracy (OA) [45], and mean intersection over union (mIoU) [46].

B. Analysis of the HSKIM Model

1) Evaluation of Hierarchical Semantic Inference Process: The HSKIM model provides a new framework to integrate hierarchical information through the iterative process of hierarchical semantic inference. To evaluate this iterative process, the energy curve of each layer is illustrated in Fig. 8. It shows that the energy curve will drop rapidly after the first iteration and converge after about three iterations. It indicates that the hierarchical semantic inference process provides a feasible way to solve the proposed model.

To further test the role of semantic inference between hierarchical layers during the iterative process, the inter-layer class relationships, $\xi(x^{(l)}, x^{(3-l)})$, are removed from the HSKIM model in this section. A comparison instance with and without $\xi(x^{(l)}, x^{(3-l)})$ is demonstrated in Fig. 9. It can be observed that the HSKIM model, with the guidance of hierarchical geographical criteria, outperforms the model without considering the inter-layer class relationships. Specifically, in the marked areas, the model without $\xi(x^{(l)}, x^{(3-l)})$ misclasses the *arbor forest* and the *shurb land* in the low layer and also fails to recognize the *farmland* and the *forest* in the high layer. It is caused that different classes in the same layer have similar appearances. For the HSKIM model, it further considers the inter-layer relationships to alleviate this problem. Namely, the HSKIM model in high layer can correct the *forest* from the *farmland* with the support of the *arbor forest* in the low layer. Moreover, the HSKIM model can address the similarity between the *arbor forest* and the *shurb land* with the guidance of the *forest* in the high layer. It indicates that semantic inference between different layers can involve more information from diverse perspectives and integrate them through the dynamic iterative process to interpret the remote sensing image. This also verifies the contribution of the HSKIM model for hierarchical segmentation.

2) Test of the Penalized Parameter q: As the low-layer classes are subclasses of the high layer's in hierarchical classes, low-layer classes belonging to the same high-layer class usually show similar appearances and bring more difficulties for semantic segmentation. To deal with this issue, the penalized parameter q is designed to distinguish these low-layer classes by increasing their penalty weight from one

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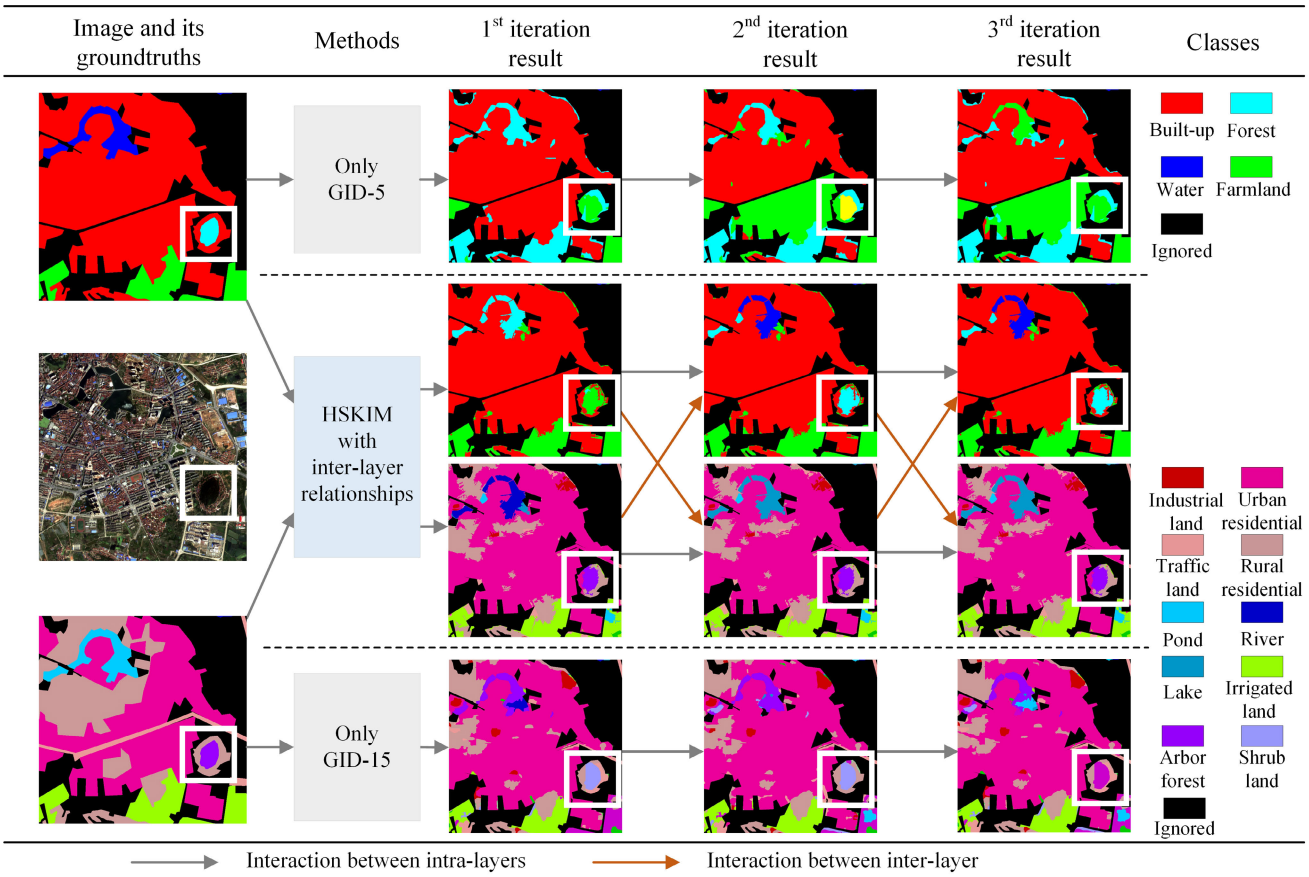


Fig. 9. Illustration of iteration results of the HSKIM model, comparing with inference only among single layer. Marked areas are used to highlight some of the classification improvements.

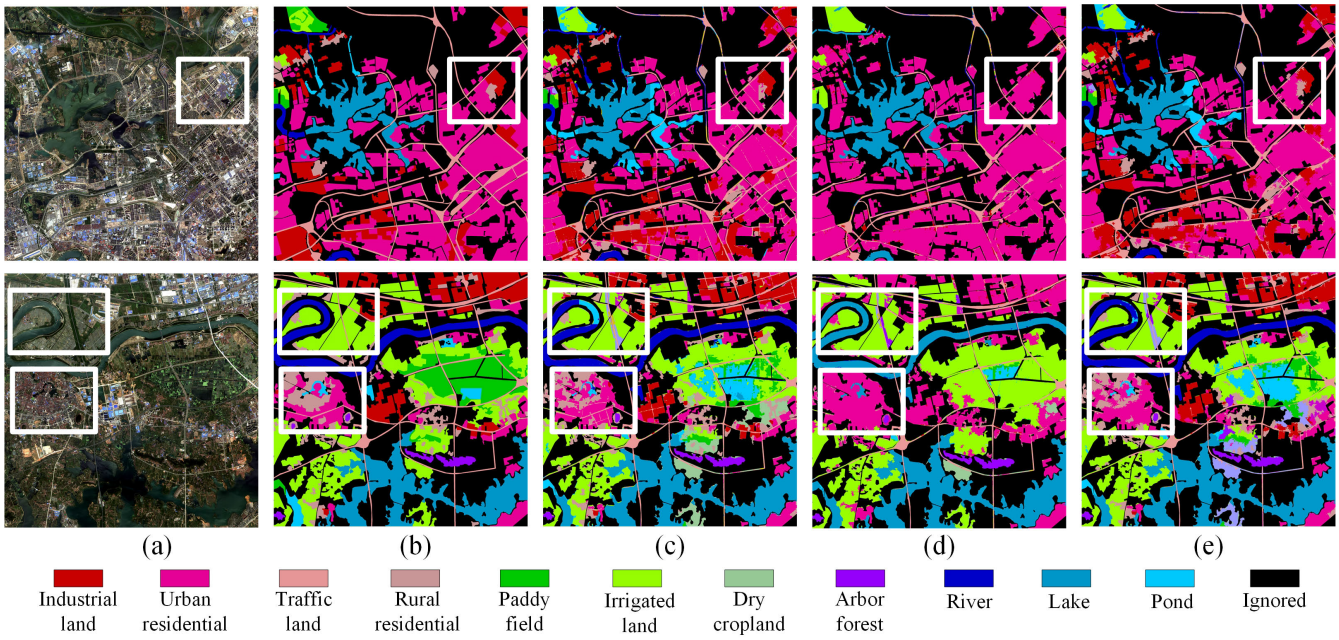


Fig. 10. Illustrations of the testing of parameter q. (a) Observed images. (b) Ground truth of the observed images on GID-15. (c) Initial results of the data learning module on GID-15. (d) Results of the HSKIM w/o q. (e) Results of the HSKIM with q. The HSKIM *w/o* q means $q = 1$ in the HSKIM model. Marked areas are used to highlight some of the classification improvements.

to $q > 1$, as given in (10). To test the role of the parameter q, the performance of the HSKIM model with q is compared with the HSKIM model without q, i.e., the penalty weight is set to one for all the situations. Experiments are implemented

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TABLE I
QUANTITATIVE RESULTS OF THE HSKIM *w/o* q AND THE HSKIM WITH q ON GID-15 (UNIT: %). THE HSKIM *w/o* q MEANS $q = 1$ IN THE HSKIM MODEL. THE HIGHEST VALUES OF EVERY METRIC ARE IN BOLD

Methods	Kappa	OA	mIoU
Baseline	71.81	77.51	51.60
HSKIM *w/o* q	66.91	73.64	41.76
HSKIM with q	**73.54**	**79.12**	**52.12**

on the ten test images of the GID dataset. Table I lists the average quantitative indicators of the HSKIM with and without q. As observed, by considering q, the Kappa coefficient, the OA, and mIoU are, respectively, improved by 1.73%, 1.61%, and 0.52% on the low-layer classes.

To further demonstrate the role of q, two experimental results are shown in Fig. 10. For the HSKIM model without q, in the marked areas of the first experiment, it misclassifies the *industrial land* and *urban residential* due to their similar appearances. Similarly, in the second experiment, the *river* and *lake*, the *lake* and *pond* are also misclassed by the HSKIM model without q. On the contrary, the HSKIM model with q effectively addresses similarities between low-layer classes within one subset, as shown in the marked regions. Table II lists the specific numerical results for each low-layer class of Fig. 10, and the HSKIM without q tends to over-classify the *urban residential* within the *built-up* subset. It brings the high indicator for over-classified *urban residential*, while the other two low indicators for underclassified *industrial land* and *rural residential*. Similar over-classification and underclassification can be observed within the subsets of *farmland* and *water*. In contrast, the HSKIM model with q can effectively resist the similarities between low-layer classes within one subset and distinguish them with higher accuracy. The outperformance shows the contribution of the penalized parameter q to the proposed HSKIM model.

3) Ablation Study: In this section, several ablation experiments are implemented to verify the contributions of three modules in the HSKIM model by using the following variants, where *w/o* denotes without.

1) *HSKIM w/o* ***D****:* Remove the CNN model from the Module ***D*** and use the unsupervised K-means algorithm to provide the initial result. By implementing this variant, it can show the impact of learning various data features from hierarchical perspectives and the impact of the initial result on the final result.
2) *HSKIM w/o* ***I****:* Remove the boundary-preservation trick from the module ***I*** and use the initial results of module ***D*** to generate hierarchical inference units. By implementing this variant, it can show the contribution of the boundary-preservation trick to the hierarchical inference unit generation.
3) *HSKIM w/o* ***S****:* Remove the inter-layer class relationship term from the Module ***S*** and implement semantic inference only within each layer. By implementing this variant, it can show the contribution of MRF-based hierarchical semantic inference.

The above ablation experiments are tested on the GID dataset. The average quantitative results of ten tested images are listed in Table III. It shows that the full HSKIM model outperforms other variants and achieves a significant improvement. Specifically, the following points illustrate the details below.

1) Compared with the HSKIM *w/o* ***D***, the full HSKIM model delivers better performance in terms of all the three quantitative indicators. It indicates that effective data feature learning and the superior initial result have an essential impact on the HSKIM model. To further illustrate it, an experimental instance is given in Fig. 11, where the HSKIM *w/o* ***D*** has misclassified fragments, such as the *farmland* and *built-up* in GID-5 and *industrial land* and *urban residential* in GID-15. But, the full HSKIM model corrects certain misclassifications by learning the hierarchical features.
2) Compared with the HSKIM *w/o* ***I***, the full HSKIM model has an improvement of all the indicators by about 3% to 5% on GID-5 and 1% to 3% on GID-15, respectively. Experiments in Fig. 11 also show that the HSKIM model can effectively recognize the boundaries of objects and generate the inference unit with clear geographical meaning by using the boundary-preservation trick.
3) Compared with the HSKIM *w/o* ***S***, the full HSKIM model, with the support of inter-layer classes, exhibits improved performance. It shows that the interaction between inter-layer classes can help the distinction of similarities between classes within the same layer.

C. Experiments of HSKIM With Different CNN-Based Methods

The HSKIM model provides a framework to realize hierarchical semantic segmentation with the inclusion of the CNN-based module ***D***. To test the effectiveness and robustness of the proposed HSKIM model against various CNN methods, 12 state-of-the-art CNN-based semantic segmentation methods are employed as the baseline for the module ***D*** of the proposed HSKIM model. They are BANet [47], BiSeNetV2 [48], DaNet [49], Deeplab V3+ [50], DnlNet [51], FCN [52], GCNet [53], KNet [54], MANet [55], Mask2former [56], Segnext [57], and UperNet [58]. Specifically, the Mask2former and Segnext are transformer-based methods, BANet, Deeplab V3+, and MANet, are proposed for semantic segmentation of the remote sensing image. Experiments are implemented on the GID dataset. The quantitative indicators are illustrated in Table IV.

As observed in Table IV, the numerical results of 12 different CNN methods show that the proposed HSKIM model can provide better performances than the baseline methods. Kappa, OA, and mIoU can be improved by 2.75%, 1.65%, 2.99% for GID-5, and 2.51%, 2.30%, 2.14% for GID-15 on average. To further illustrate the advantages of the proposed model, an instance of 12 different baseline methods and their corresponding results with the HSKIM model are shown in

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TABLE II

QUANTITATIVE RESULTS OF THE TESTED IMAGES ILLUSTRATED IN FIG. 10 (UNIT: %). IMAGE_1 AND IMAGE_2 ARE THE IMAGES SHOWN IN FIG. 10(A). THE HSKIM *w/o q* MEANS $q = 1$ IN THE HSKIM MODEL. THE HIGHEST VALUES OF EVERY METRIC ARE IN BOLD

	Methods	*Ind. land (Built-up)*	*Urb. resid. (Built-up)*	*Rur. resid. (Built-up)*	*Traff. land (Built-up)*	*P. field (Farmland)*	*Irr. land (Farmland)*	*River (Water)*	*Lake (Water)*	*Pond (Water)*	*Arb.forest (Forest)*	Kappa	OA	mIoU
Image_1	Baseline	80.77	76.78	68.95	68.83	24.54	76.85	92.19	96.99	34.35	-	71.54	74.35	60.11
	HSKIM w/o q	3.48	**85.51**	8.12	62.56	3.65	88.33	1.79	**99.09**	0.28	-	51.91	56.90	30.61
	HSKIM with q	**84.39**	80.57	**69.78**	**69.18**	**24.58**	**88.74**	**98.07**	98.77	**42.67**	-	**73.58**	**76.32**	**60.86**
Image_2	Baseline	81.19	83.87	72.20	71.20	24.37	81.14	67.72	71.32	5.34	90.23	71.55	78.58	47.71
	HSKIM w/o q	5.37	**97.26**	6.78	65.80	0.26	**95.85**	2.26	**99.05**	0.16	94.80	62.69	74.27	30.30
	HSKIM with q	**82.33**	86.51	**75.34**	**72.94**	**25.31**	82.97	**71.34**	74.48	**8.24**	**94.83**	**74.19**	**80.95**	**50.59**

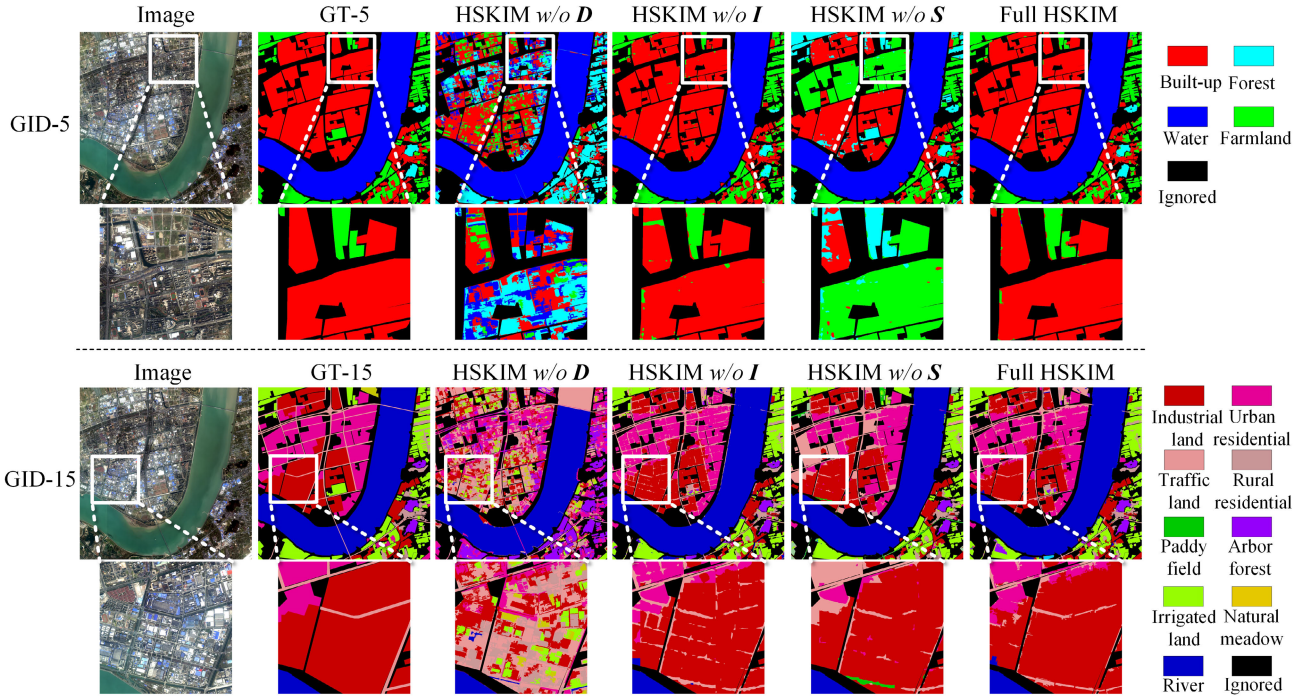


Fig. 11. Illustration of the ablation study on GID-5 (up) and GID-15 (down). The ground truth of GID-5 and GID-15 is abbreviated as GT-5 and GT-15.

TABLE III

ABLATION STUDY ON GID DATASET (UNIT: %). THE HIGHEST VALUES OF EVERY METRIC ARE IN BOLD

Dataset	Module *D*	Module *I*	Module *S*	Kappa	OA	mIoU
GID-5	–	✓	✓	66.37	72.97	47.52
	✓	–	✓	89.73	93.58	79.94
	✓	✓	–	88.82	93.09	76.94
	✓	✓	✓	**92.78**	**96.07**	**84.92**
GID-15	–	✓	✓	45.14	50.09	31.47
	✓	–	✓	71.46	78.79	50.24
	✓	✓	–	69.83	77.28	48.93
	✓	✓	✓	**74.15**	**79.48**	**52.66**

Fig. 12. For experiments of GID-5, the baseline methods can effectively extract features of each class by learning from the training data. However, different land use or land cover classes may have similar appearances. It will lead to misclassifications between these classes, such as the misclassifications between the *built-up* and *farmland* in Fig. 12. In contrast, the proposed HSKIM model not only considers data features learned by the baseline but also integrates geographical relationships between hierarchical classes. Hence, the HSKIM model performs better than the baseline. Moreover, due to the increased number of classes in GID-15, low-layer classes belonging to the same high-layer class are prone to have similar appearances, making the baseline more challenging to recognize classes in GID-15. Since the geographical relationships self-learned within the low layer are guided by the high layer in the HSKIM model and the penalized parameter q, it can further improve the performance of the baseline, especially in distinguishing low-layer classes belonging to the same high-layer class. As observed in the recognition of the *river* in Fig. 12, most of the baseline methods cannot distinguish the *river* and *lake*, but

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TABLE IV
QUANTITATIVE COMPARISON RESULTS OF 12 STATE-OF-THE-ART CNN METHODS AND THE HSKIM MODEL BASED ON THE GID DATASET (UNIT: %). THE HIGHEST VALUES OF EVERY METRIC ARE IN BOLD

Methods	GID-5						GID-15					
	Baseline			HSKIM			Baseline			HSKIM		
	Kappa	OA	mIoU	Kappa	OA	mIoU	Kappa	OA	mIoU	Kappa	OA	mIoU
BANet [47]	89.60	94.48	78.28	**92.01**	**95.76**	**82.63**	74.36	80.09	55.13	**76.84**	**82.32**	**56.86**
BiSeNetV2 [48]	90.88	95.17	81.00	**92.32**	**95.98**	**81.85**	73.17	78.70	52.10	**74.51**	**79.95**	**54.09**
DANet [49]	92.02	95.73	83.72	**94.44**	**97.11**	**86.23**	76.63	81.49	52.24	**79.07**	**83.60**	**53.18**
DeepLab V3+ [50]	85.48	91.53	74.87	**91.44**	**95.42**	**80.99**	72.93	78.52	48.54	**75.70**	**80.94**	**53.08**
DNLNet [51]	88.13	93.02	80.32	**92.78**	**96.07**	**84.92**	71.97	77.66	49.35	**74.15**	**79.48**	**52.66**
FCN [52]	86.68	92.37	74.69	**91.50**	**95.43**	**80.04**	72.15	77.60	49.81	**74.45**	**79.67**	**49.98**
GCNet [53]	90.74	94.92	80.88	**93.66**	**96.64**	**84.39**	70.36	76.40	47.89	**73.25**	**78.89**	**51.70**
KNet [54]	94.27	97.06	86.35	**95.26**	**97.61**	**86.92**	71.81	77.51	51.60	**73.54**	**79.12**	**52.12**
MANet [55]	93.27	96.44	86.06	**94.60**	**97.15**	**89.17**	74.28	79.09	54.22	**76.21**	**80.86**	**58.24**
Mask2former [56]	90.20	94.85	82.88	**93.26**	**96.46**	**84.95**	73.49	78.49	53.39	**75.64**	**81.06**	**53.93**
Segnext [57]	95.19	97.52	88.00	**95.90**	**97.89**	**89.02**	79.83	83.98	57.81	**81.77**	**85.66**	**60.10**
UperNet [58]	91.51	95.41	82.71	**93.78**	**96.77**	**84.55**	74.07	75.96	47.52	**79.96**	**81.60**	**49.38**

the HSKIM model performs well. By the above comparison results and discussion, it can verify the effectiveness and robustness of the proposed HSKIM model against different CNN-based methods.

D. Comparison Results With MRF-Based Methods

In the HSKIM model, a new MRF-based model is developed to realize semantic inference between intra-layer and inter-layer by self-learning hierarchical knowledge relationships. To test its effectiveness, other four MRF-based methods are employed in this section for comparison, as follows.

1) *Iterated conditional mode (ICM) [17]:* It is a classic pixel-based MRF model that sequentially maximizes the local conditional probability to get the final segmentation. It can infer the labels of pixels along their edges at the pixel layer.
2) *Object-based MRF model with generative probability inference (OMRF) [9]:* It is a classic object-based MRF model that treats the superpixels as the objects and uses them as the basic vertexes for the probability graph. It infers the relationships between adjacent objects and provides the segmentation result by maximizing the posterior marginal.
3) *Hybrid granularities MRF model with auxiliary label fields (HMRF-A) [38]:* It is a hybrid MRF model built on both pixel and object granularities, which provides a way to realize inter-layer semantic inference. However, the relationship between inter-layer labels is constructed without specific geographical criteria.
4) *Self-learning-update CNN Model (SLU-CNN) [21]*: The SLU-CNN model is a data–knowledge dual-driven model. It proposes a self-learning trick to identify each object from the input image with the CNN method and uses the OMRF model to realize semantic inference. But, it is implemented on only one layer.

Comparing the HSKIM model with the above methods can reveal the difference between MRF-based inference within a single layer, such as the pixel-based layer (ICM) or the object-based layer (OMRF and SLU-CNN), and the hierarchical layer (HMRF-A and HSKIM). Additionally, it can also demonstrate variations between semantic inference without geographical criteria (ICM, OMRF, and HMRF-A) and with geographical criteria (SLU-CNN and HSKIM). These methods are tested on the GID dataset as well. To implement a robust comparison with these MRF-based methods, DeepLab V3+, BANet, and MANet, proposed for semantic segmentation of the remote sensing images, are employed as baseline methods in this section. To ensure a fair evaluation, the hyperparameter settings for each method align with the authors' tests of their respective models.

As illustrated in Table V, segmentation results of the DeepLab V3+ model are employed as the initial results for these MRF-based methods to demonstrate that the HSKIM model outperforms the other four MRF-based methods on both GID-5 and GID-15. A GID instance is given to further demonstrate performances of the above MRF-based methods and the HSKIM model in Fig. 13 as well. In the experiment results, the ICM model has many fragment misclassifications due to the pixel-based limitations, such as the *farmland* of GID-5 and the *river* of GID-15. The OMRF model can provide a smoother result by considering the relationship between objects, but still has wrong recognition as it is a single layer model, such as over-classified the *water* of GID-5 and underclassified the *irrigated land* of GID-15. The HMRF-A model proposes a hybrid model to integrate the pixels and the objects. Hence, it shows a better performance than the previous two models, such as the recognition of the *farmland* of GID-5,

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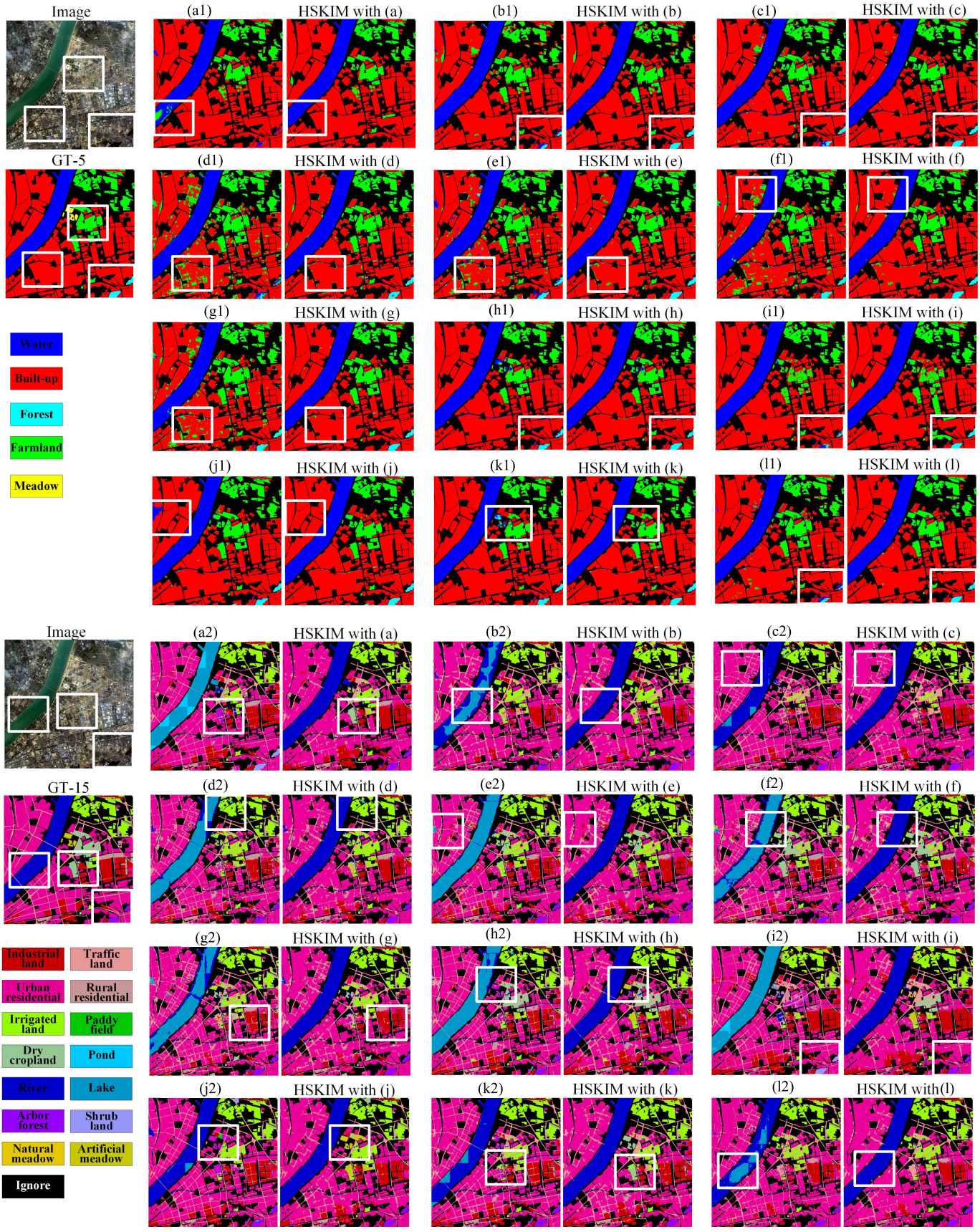


Fig. 12. Illustration of comparison results between different CNN methods and the HSKIM model with these methods on GID-5 and GID-15. The ground truth of GID-5 and GID-15 is abbreviated as GT-5 and GT-15. (a1) and (a2) BANet. (b1) and (b2) BiSeNetV2. (c1) and (c2) DANet. (d1) and (d2) DeepLab V3+. (e1) and (e2) DNLNet. (f1) and (f2) FCN. (g1) and (g2) GCNet. (h1) and (h2) KNet. (i1) and (i2) MANet. (j1) and (j2) Mask2former. (k1) and (k2) Segnext. (l1) and (l2) UperNet.

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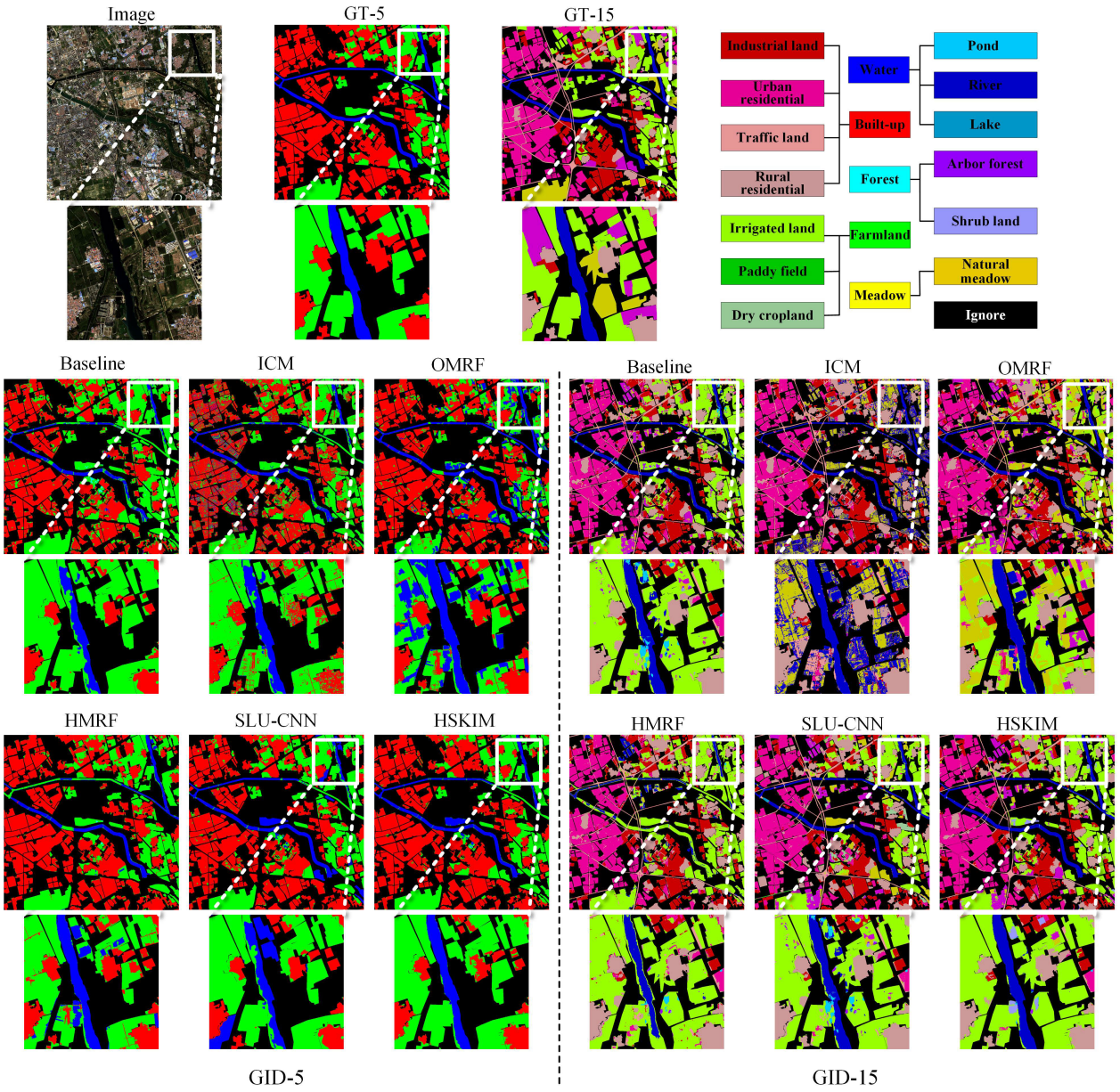


Fig. 13. Illustration of comparison results between the ICM, OMRF, HMRF-A, SLU-CNN, and the proposed HSKIM model on GID-5 (left) and GID-15 (right). The ground truth of GID-5 and GID-15 is abbreviated as GT-5 and GT-15.

and the *irrigated land* of GID-15. But, it suffers the lack of the geographical knowledge and confuses the *water* to *farmland* in the high layer, even if the classes belonging to its subset are correctly segmented into the *irrigated land* in the low layer. By capturing the knowledge relationship from the initial CNN result, the SLU-CNN model further improves the accuracy, especially in the *farmland* of GID-5 and *river* of GID-15. However, its knowledge is constructed only on the single layer without geographical relationships between hierarchical layers. This leads to the misclassifications between the *irrigated land* and *pond* belonging to different subsets in the high layer. Different from the above MRF-based methods, the HSKIM model considers hierarchical knowledge relationships under the geographical criteria and designs a self-learning MRF-based model to implement semantic inference. Hence, the HSKIM model performs better than other MRF-based methods regarding semantic inference and hierarchical information integration.

E. Complexity Analysis

The proposed method provides a framework to integrate data features and hierarchical semantic inference among its three modules. Since the computational time of module $\boldsymbol{D}$ depends on the selected CNN methods, the complexity analysis mainly focuses on modules $\boldsymbol{I}$ and $\boldsymbol{S}$. The computational complexity of modules $\boldsymbol{I}$ and $\boldsymbol{S}$ is $O(KNT)$, where K is the number of classes, N is the number of inference units or over-segmented regions, and T is the number of iterations. To further analyze the complexity of these methods, the

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TABLE V
QUANTITATIVE INDICATORS OF ICM, OMRF, HMRF-A, SLU-CNN, AND HSKIM IN EXPERIMENTS OF GID DATASET (UNIT: %). THE HIGHEST VALUES OF EVERY METRIC ARE IN BOLD

	Index	GID-5						GID-15					
		Baseline	ICM	OMRF	HMRF	SLU-CNN	HSKIM	Baseline	ICM	OMRF	HMRF	SLU-CNN	HSKIM
DeepLab V3+	Kappa	85.48	80.92	81.78	86.59	87.88	**91.44**	72.93	52.13	69.54	70.44	70.80	**75.70**
	OA	91.53	88.97	89.71	91.24	92.58	**95.42**	78.52	59.48	75.97	76.41	75.15	**80.94**
	mIou	74.87	68.84	71.99	75.98	76.67	**80.99**	48.54	32.17	47.47	45.24	48.24	**53.08**
BANet	Kappa	89.60	81.01	86.93	89.98	90.36	**92.01**	74.36	52.73	68.03	71.36	72.94	**76.84**
	OA	94.48	88.56	92.07	94.51	94.92	**95.76**	80.09	59.94	74.65	77.84	78.59	**82.32**
	mIou	78.28	70 .82	75.32	78.39	79.28	**82.63**	55.13	32.90	50.96	52.08	52.73	**56.86**
MANet	Kappa	93.27	83.69	89.01	94.07	94.58	**96.40**	74.28	53.31	68.03	71.36	72.94	**76.21**
	OA	96.44	89.33	92.46	96.53	96.79	**97.15**	79.09	58.51	74.65	77.84	78.59	**80.86**
	mIoU	86.06	73.47	80.31	86.15	87.21	**89.17**	54.22	35.06	50.96	52.08	52.73	**58.24**

TABLE VI
COMPUTATIONAL TIME OF DIFFERENT MRF-BASED METHODS ON THE GID DATASET. HERE, t_0 IS THE TIME OF THE SELECTED CNN MODEL IN THE DATA LEARNING MODULE

Dataset	ICM	OMRF	HMRF	SLU-CNN	HSKIM
GID-5	t_0+251.99s	t_0+95.60s	-	t_0+58.46s	-
GID15	t_0+310.71s	t_0+111.46s	-	t_0+98.24s	-
Total	$2t_0$+562.70	$2t_0$+197.06s	$2t_0$+317.40s	$2t_0$+156.70s	$2t_0$+219.02s

average computational times of test images in the GID dataset are illustrated in Table VI, where the time of module $\boldsymbol{D}$ is denoted as t_0. By comparing with other MRF-based methods, the proposed HSKIM shows a faster speed compared to the pixel-based ICM method and a similar performance to the other OMRF-based methods.

V. CONCLUSION

In this article, an HSKIM model is proposed to implement semantic inference among hierarchical classes and integrate their data features for hierarchical semantic segmentation of remote sensing images. The strength of the proposed HSKIM model mainly lies in three aspects. The first is that the HSKIM model presents a new framework to extract geographical objects and implement hierarchical semantic inference guided by geographical criteria. The second is that the HSKIM model designs a new boundary preservation trick based on GEOBIA to effectively extract geographical objects as the inference units. Last but not least, the HSKIM model develops a new MRF-based model to dynamically integrate data features and geographical relationships between both intra-layer and inter-layer by a self-learning technique. Experimental results, employing 12 state-of-the-art CNN-based methods as the baseline and comparison with four MRF-based methods on the GID dataset, show the robustness and effectiveness of the proposed HSKIM model.

The proposed framework enables hierarchical semantic inference based on specific geographical classification criteria. In future work, we will extend the HSKIM model by incorporating additional diverse geographical criteria or integrating domain-specific expertise to further refine and guide the semantic inference process of remote sensing images.

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